

Learning to Ask: Adaptive Query Strategies for Interactive Song Identification across Noise and clean conditions

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Abstract

This paper presents a unique interactive song identification system that analyses multiple AI approaches to find the song that the user is thinking of through a series of yes-or-no questions. The target is to find and analyse existing methods for optimal question selection to reduce the state space. The system evaluates three unique approaches here: an entropy-based hard elimination model, an ML-based probabilistic model and an innovative adaptive engine that balances between exploration and exploitation. To imitate real world situations, a noise simulation mechanism is integrated into the system to test the models under various degrees of noise. Experimental tests show that under noise-free conditions, all three models perform with high accuracy, with the entropy model edging the others in convergence rate and average number of questions. However, it collapses under the introduction of noise. The paper attempts to provide alternative methods to cope with and combat the fall of accuracy under the introduction of noise by exploring different methods. The ML model is thus explored. It gives good accuracy even under noise, but its rigid, pretrained structure limits its flexibility and dynamic nature. To address this, we propose an innovative adaptive engine that adjusts the question to ask based on its current state, confidence and uncertainty while retaining the influence and information of the ML model. Under different noise levels, the adaptive model consistently outstrips the other two in an overall engine score, which is calculated through the accuracy and rate of convergence. These results and inferences show the importance of noise-aware design to mimic real life and establish an adaptive questioning strategy as an optimal strategy for real-world interactive AI systems/

Keywords— Interactive AI, Decision-Making Systems, Adaptive Questioning, Exploration–Exploitation Tradeoff, Entropy-Based Methods, Machine Learning, Noise Robustness

I. INTRODUCTION

Interactive systems that depend on choosing optimal questioning strategy to minimize the state space in an optimal manner has been becoming increasingly relevant in various domains like recommendation systems, information retrieval, and decision recommendation systems, information retrieval, and decision support.

However, one of the critical assumptions made in a majority of the systems is that the user gives correct data or answers. However, in real world it is rarely the case. Users may provide incorrect, uncertain, or inconsistent answers because of ambiguity, lack of

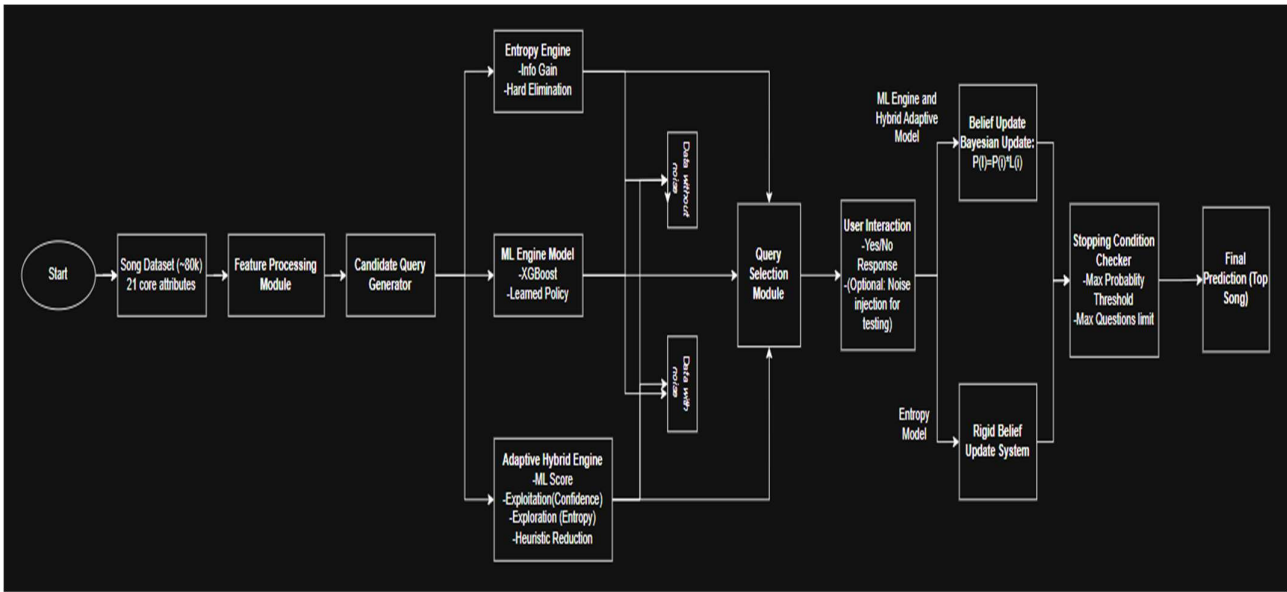
knowledge, or subjective interpretation of questions. Such noise in human input can significantly degrade system performance, particularly in methods that rely on deterministic elimination strategies. Among the existing approaches, entropy-based systems are widely regarded as the most efficient under ideal conditions without any noise, as it selects the questions that maximise the information gain and implement hard elimination of questions. However, this can prove to be brittle, as a single incorrect response can overthrow several prior observations. In contrast to this, machine learning-based soft belief systems tend to perform better under noise, calibrating their probability based on the questions. However, these techniques often use static or pre-fixed strategies and lack dynamic adaptation as per the current state space and uncertainty levels.

This limitation paves the need for adaptive decision- making techniques that can function optimally under noise, which resembles real-life uncertain or ambiguous user interactions. Thus, in this work, we propose an adaptive questioning engine that balances exploration and exploitation techniques during the question selection process. The system dynamically adjusts its questioning strategy and interleaves between different approaches based on the current belief state and the uncertainty levels, enabling it to reduce the impact of noisy responses while ensuring optimal convergence rate. To further ensure realistic evaluation, we introduce a noise simulation framework that models user errors during interaction, allowing systematic analysis of system performance under varying levels of uncertainty. Through this framework, we conduct a comprehensive comparative study of entropy-based, probabilistic, and adaptive approaches.

The main contributions of this work are as follows:

1. Comparison between the performances of different models in choosing the right questions under uncertainty and noise and an innovative new approach for it.
2. A noise simulation model to mimic real-life scenarios and analyse the model performance under noise.

An extensive experimental evaluation showing the improved accuracy and convergence under the proposed adaptive engine strategy.



2. LITERATURE REVIEW

(Gal et al., 2023) have analyzed question-asking strategies using the Akinator game through a data-driven approach. The study applies topic modelling techniques such as LDA and BERT to identify patterns in questioning behaviour and highlight similarities with human information-seeking processes. However, the approach relies on reverse engineering due to a lack of access to the underlying algorithm, limiting the depth of the analysis.

(Guzmán Colin, 2023) proposes an Akinator-inspired framework to improve human–AI interaction by shifting from direct answer generation to iterative and adaptive questioning. The model treats AI as a cognitive support system that progressively refines user intent and improves decision-making. It enhances communication by reducing ambiguity through structured inquiry. However, the approach is primarily conceptual and lacks empirical validation in practical scenarios.

(Yung et al., 2022) have proposed a Bayesian-based troubleshooting system that uses sequential questioning to identify errors and mistakes under uncertain conditions. The system also showcases robustness to incorrect and wrong answers and is validated in an industrial environment. However, the performance depends on the quality and quantity of questions and knowledge of the base design.

(Sasson Lazovsky et al., 2024) explore the relationship between question-asking and prompt engineering, showcasing the shared dependence on creativity, clarity, and critical thinking. The study shows that both processes are important to reduce uncertainty and improve communication between humans and AI systems. However, currently the work is mainly theoretical and has not been provided with any proper practical implementation and system to verify it.

(Whang et al. (2013)) propose selecting questions that maximise expected accuracy in crowd-assisted entity resolution using probabilistic modelling. This method

reduces the number of queries but on the other hand requires approximation due to high computational complexity. It is effective but mainly limited to pairwise matching scenarios.

(Berger-Tal et al. (2014)) propose a dynamic model where the exploration–exploitation balance evolves over time, leading to four phases: establishment, accumulation, maintenance, and exploitation. The framework is broadly applicable across domains and offers strong conceptual insights. However, it is highly theoretical, relies on simplifying assumptions, and lacks strong empirical validation

De Bruyn et al. (2022) use the Twenty Questions game to evaluate world knowledge in language models through yes/no question answering. They find that large models (e.g., GPT-3) perform well but struggle with size and shape reasoning, while smaller models show inconsistent knowledge. They improve smaller models using external knowledge grounding (e.g., Wikipedia). However, the setup relies on simplified yes/no queries and predefined objects, limiting real-world applicability.

Zhang et al. (2025) propose Entro-duction, which dynamically adjusts LLM reasoning depth using entropy-based uncertainty signals. It improves accuracy and efficiency by avoiding over/under-reasoning. However, it relies on sensitive entropy estimates and lacks broad real-world validation.

3. Proposed Methodology

3.1 System Overview

The proposed system is an interactive decision-making framework similar to the Akenator game, designed for song identification through a sequence of binary (yes or no) questions. The system functions on a large dataset of 80k songs, each represented by 21 discrete attributes such as genre, artist name, year, etc. These features form a

structured attribute space from which the candidate questions are generated and evaluated during the process.

At its crux, the system has a dynamic candidate space associated with a probabilistic belief distribution over all possible songs. Initially, each candidate has an equal predefined probability. At each step, the system constructs all possible questions that can be asked, evaluates them with a scoring mechanism and the most informative query is selected to be asked. Based on the user's response the internal system state is updated.

The framework compares three distinct decision-making systems: an entropy-based model, a machine learning-based model, and a proposed adaptive questioning engine. The entropy-based model uses hard elimination by selecting the questions that maximize the question gain and implements rigid reduction of states. The machine learning model, on the other hand, employs a trained XGBoost model to estimate the effectiveness of various questions and choose the best one, effectively reducing the state space in an efficient manner using soft weighted elimination. The adaptive method further extends the ideas of both models, including different approaches of investigating the state space, namely exploration and exploitation, integrating confidence-based heuristics and information-theoretic measures within a comprehensive scoring framework.

The machine learning component is trained using simulated questions, where the effectiveness of each function is mapped based on its ability to reduce the candidate space. This allows the model to assemble the states and interaction context into a single cohesive framework. Building upon this, the adaptive engine includes real-time feedback from the changing belief distribution, enabling it to dynamically adjust its questioning strategy. Specifically, the system is thus equipped to switch between exploration and exploitation techniques during the system run, by modulating the influence of the various scoring components based on the current level of uncertainty.

The questions are asked iteratively, each time the system adjusting and calibrating its belief distribution, and the reevaluated candidate space influencing the selection of subsequent questions. This continues until a stopping condition is reached, either achieving a high confidence prediction or reaching the predefined question limit. The final answer prediction is determined by selecting the candidate with the highest probability and confidence.

3.2. Entropy-Based Model

The first approach that was explored was an entropy-based model with rigid elimination. It operates under the assumption that the most optimised question that can be chosen at any moment is the one that maximises the expected reduction in the candidate space.

At each iteration, a set of the remaining candidate songs is maintained. All the possible feature-value pairs are considered for potential questions. For any given state feature query the remaining set is split into two possible subsets. To maximise the information gain and reduction in space, queries which produce more balanced splits tend to be preferred.

Apart from the standard information-theoretic calculation, three other components are integrated into the question selection. These are

- (i) reduction potential, which captures the amount of the candidate space eliminated by the query,
- (ii) balance, which reflects how evenly the candidate set is partitioned,
- (iii) confidence, which measures the decisiveness of the split.

These components are incorporated using a weighted formulation to provide a comprehensive score for each question.

In order to ensure the same feature is not used for questioning, a category-based penalty is applied, calibrating the weights to avoid it. The query with the highest overall score after this is selected and asked to the user. Based on the user response, the candidate space is updated through hard elimination. This rigid elimination is done assuming noise-free ideal answers.

This process is repeated iteratively until the candidate space is sufficiently reduced or a stopping condition is met. While this approach is highly effective in noise-free environments and often produces the correct output in a short number of queries, it is extremely sensitive to noise and tends to break even at the occurrence of one inaccuracy or error.

Thus, this system is not suitable in real-life systems where cases of uncertainty and ambiguity might affect the accuracy of data being given. The equations are an exception to the prescribed specifications of this template.

3.3. Machine Learning-Based Model

The second approach departs from rule-based question selection and instead formulates the problem as a learned decision process. Rather than relying on fixed heuristics, the system is trained to estimate the utility of candidate queries by leveraging patterns extracted from simulated interactions. This change causes the model to capture relationships and identify optimal choices that can be skipped or glossed over by manual method. The overall method has several stages as follows:

3.3.1. Dataset Preparation and Feature Representation

The system operates on a dataset comprising approximately 80,000 songs, each characterised by 21 discrete attributes derived from audio features and associated metadata. These attributes include categories like genre, mood, tempo category, language, and other high-level audio properties. Continuous variables are categorised and calibrated into discrete values to ensure consistency in representing queries as feature-value pairs and ensures proper model inclusion and functioning.

3.3.2. Simulation of Interaction Data

To construct the data for training, the actual process is simulated repeatedly under various conditions. In each run, a target song is selected at random, and the system takes up all the feasible feature-value pairs to ask questions from. T

The utility of these questions is then assessed by examining the effect on the size of the candidate space. It is assessed mainly by the reduction in the candidates after the enquiry.

$$S_{target} = \frac{N_{before} - N_{after}}{N_{before}} \quad (1)$$

where N_{before} and N_{after} denote the number of candidates prior to and following the application of the query, respectively. This formulation provides a direct measure of how informative a question is, with higher values indicating better differentiative ability.

3.3.3. FEATURE ENCODING FOR LEARNING

Each training set takes both the state of the system and the question that is being used. The state is described by variables such as the number of remaining candidates, the current entropy, the frequency with which the given category has been used, etc. The question is encoded using a one-hot representation of the feature-value pair. This ensures the ability of the model to differentiate between queries and maintain a consistent input structure.

3.3.4. Model Training

An XGBoost model is trained on the dataset to infer the relationships and approximate the mapping between the system state, the query selection, and expected utility. By this, the model learns to predict how useful a question is likely to be given the current state and context. Once trained, it functions as a scoring mechanism capable of predicting scenarios beyond just the ones it was trained on.

3.3.5. Runtime Question Selection

During testing and inference, the model is loaded and made to evaluate candidate questions at each step of the process. For every key-value pair, an input of either yes or no is considered and passed to the model to get a predicted score.

However, reliance on learned score alone can lead to highly skewed partitions. TO reduce this, a balance term is introduced, reflecting the degree to which a query divides the candidate space evenly. The scoring function is given as

$$S(q) = \alpha \cdot S_{ML}(q) + (1 - \alpha) \cdot S_{balance}(q) \quad (2)$$

where $S_{ML}(q)$ denotes the model-predicted score, and the balance component is defined as:

$$S_{balance} = 1 - |P_{yes} - P_{no}| \quad (3)$$

Here, P_{yes} and P_{no} represent the proportions of candidates satisfying and not satisfying the query, respectively. The parameter α governs the relative influence of the learned model. The query that optimises and maximises this combined score is selected for interaction.

3.3.6. Probabilistic Belief Update

In contrast to the entropy-based model, which eliminates the candidate aggressively if it doesn't satisfy the given constraints, this model uses a soft updating and deletion probabilistic model. After each user response, the belief of that state is updated according to a likelihood-based rule:

$$P_i^{(t+1)} \propto P_i^{(t)} \cdot L_i \quad (4)$$

where $P_i^{(t)}$ IN (4) denotes the probability assigned to the candidate i at step t , and L_i corresponds to the likelihood of observing the given response under that candidate. This formula allows the system to have multiple plausible hypotheses, thus reducing its impeding sensitivity to occasional errors or inconsistencies from user.

3.3.7. Termination Criteria

This process continues iteratively, with each step modifying and refining the belief distribution and probability and the questions being influenced by a new set of state every time. The process ends when the system either achieves a sufficiently high confidence on a single state or the maximum query limit is reached. The final predictions is thus obtained by selecting the candidate with the highest probability/

3.4. Adaptive Hybrid Questioning Model

This method builds upon the previous two models by combining a model-based learned decision policy with a dynamically altering questioning strategy. Rather than just relying on either heuristic values or a fixed learned model, this approach makes use of both of them to make decisions, leveraging their data to predict the compatibility of a question with the current state to provide an optimal output. The key point of the system is the change in making the utility of a question not absolute, but contingent on the system's current certainty level. The model is designed to respond fluidly and alternate between exploratory and exploiting phases cued by various degrees of uncertainty during the interaction.

3.4.1. OVERALL FRAMEWORK

The system uses the same dataset and feature representation as used in the prior approaches, maintaining a probabilistic distribution over the candidate space. At every run, the task is to select a query from the feasible key-value space. What differentiates this approach from other models is not the state space, but the way in which it is evaluated and chosen upon. The decision is no longer just due to one barrier or reason. Rather, the decision is taken as a culmination of several signals, each representing a different aspect of the query's usefulness.

3.4.2. Multi-Component Query Evaluation

For each query q , the model has a set of scoring components. By amalgamating all of them, we are able to get a more accurate assessment than any independent single variable can produce.

An estimate is first obtained from the trained regression model as:

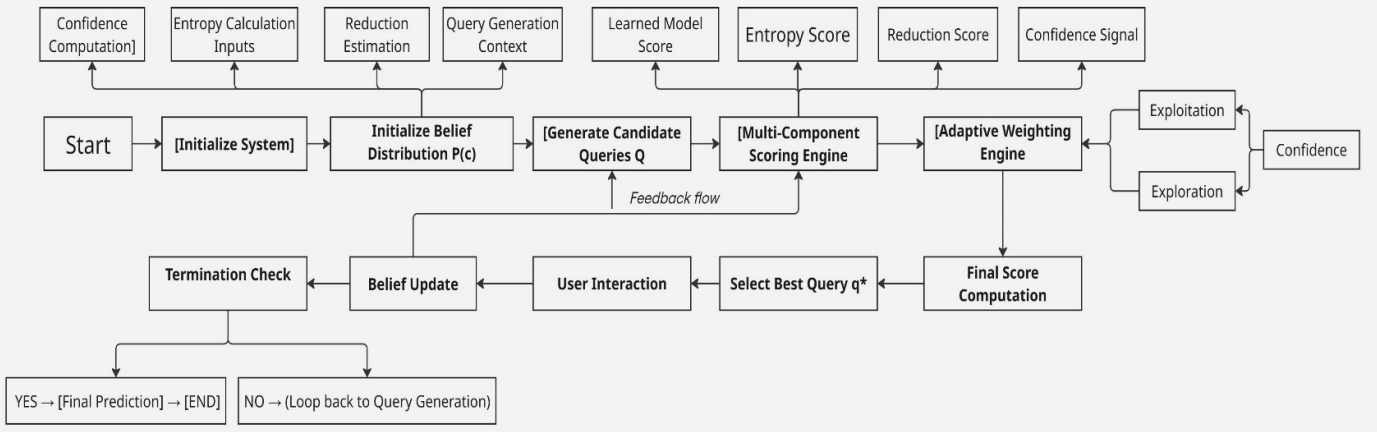
$$S_{ML}(q) = \text{XGBoost}(x_q) \quad (5)$$

where x_q denotes the feature vector encoding both the system state and the query. This component reflects patterns inferred from prior simulated interactions and can be interpreted as a prior over query effectiveness.

However, just using the learned model predictions is not enough. This is particularly true for states that deviate from the training distribution. This leads to the introduction of a new term, defined through entropy-based approximation as:

$$S_{explore}(q) = -P_{yes} \log P_{yes} - P_{no} \log P_{no} \quad (6)$$

Here, P_{yes} and P_{no} correspond to the proportions of candidates satisfying and not satisfying the query. Usually, questions that induce a more balanced 50:50 splits tend to be favoured by these, thus encouraging a uniform split commonly termed as exploration.



In order to counteract or contrast it, an exploiting parameter is also used to sometimes take more skewed splits if it ensures a smaller state space after computation. Let:

$$C = \max_i P_i$$

represent the maximum posterior probability across candidates. This quantity serves as a proxy for certainty. As C values gradually increases, it becomes better to refine the more probable candidates rather than just exploring the space indiscriminately.

In addition, a heuristic reduction term is incorporated to retain sensitivity to elimination dynamics:

$$S_{heuristic}(q) = \frac{N_{before} - \min(N_{yes}, N_{no})}{N_{before}} \quad (7)$$

This shows the extent in which the candidate space is contracted by a query without enforcing rigid elimination.

3.4.3. Adaptive Exploration–Exploitation Balancing

An important quality of the model is how these components are not just split into rigid divisions to come into play in fixed phases, but the system itself continuously modulates its behaviour through confidence-dependent weights.

$$w_{explore} = 1 - C, w_{exploit} = C \quad (8)$$

Thus, this results in the encoding of a soft transition between exploratory and exploitative phases. When there is high uncertainty, that is, C is small, the system privileges information gain. On the other hand as confidence increases, the preference shifts to refining the leading hypothesis. The transition is gradual, and no fixed phase boundary is present.

3.4.4. Final Scoring Function

The final utility score of a function is obtained by aggregating all the individual components that we have calculated by various methods until now. It is given by:

$$S(q) = w_1 \cdot S_{ML}(q) + w_2 \cdot w_{explore} \cdot S_{explore}(q) + w_3 \cdot w_{exploit} \cdot S_{exploit}(q) + w_4 \cdot S_{heuristic}(q) \quad (9)$$

where w_1, w_2, w_3, w_4 are fixed coefficients that regulate the relative contribution of each term. The selected query is simply the one that maximises $S(q)$. While the formula seems simple and straightforward, the actual computation is sensitive due to gradual increase in confidence and a shift between exploratory and exploitative phases.

3.4.5. Probabilistic Belief Update

After each interaction, the belief distribution is updated once again through a likelihood-based mechanism:

$$P_i^{(t+1)} \propto P_i^{(t)} \cdot L_i \quad (10)$$

where $P_i^{(t)}$ denotes the probability of candidate i at step t , and L_i represents the likelihood of the observed response under that candidate. This update ensures we do not cause any irreversible elimination, therefore allowing the system to have alternative hypotheses even in the presence of ambiguity and external noise.

3.4.6. Iterative Decision Process

This process progresses iteratively. With each new question, the belief state is altered, and the scoring is recalibrated accordingly. Termination is triggered once either one candidate predominates or the query limit has been exceeded. Then, upon the termination condition, the candidate with the highest probability is selected and displayed.

3.4.7. Discussion

The model is a hybrid between a policy learning model and adaptive control based on entropy and dynamic search space optimisation. This dual mode structure is extremely advantageous under uncertainty or noise. When purely deterministic models risk failure, static learning models lack responsiveness, the currently proposed model brings the best out of both worlds and gives a balance and resilient behaviour.

4. Experimental Setup

The system is checked and evaluated under a controlled simulation framework that mimics interactive question

answering sessions under both ideal and noisy conditions. The whole process ranges from integrating the dataset preparation, feature representation, parameterised simulation to final comparative analysis between the models.

4.1. Dataset and Feature Representation

Experiments are conducted using a dataset of approximately 80k songs, each having 21 core attributes derived from audio characteristics and metadata. These include attributes like genre, language, artist name, tempo, energy, danceability, etc.

To support binary querying, all continuous features are discretised into categorical levels. This representation preserves sufficient granularity for discrimination while maintaining interpretability within the question-answer framework.

4.2. Simulation Framework

Evaluation and scoring is done by a framework that simulates sequential questioning like a real human-taker interaction. In each run a target song is selected randomly and chosen to be the one thought by the user.

Main parameters include the number of runs, dataset size, maximum number of questions and noise level. Each individual run functions on a random subset of songs as per the given user input.

At each step, a query is chosen as per the most optimal approach, and it receives an output which is used to update the belief distribution. This repeats until a stopping condition is met.

4.3. Noise Modelling

In order to ensure the model works well in real life condition noise is introduced in the responses. For a given noise level η , responses are flipped at regular intervals (every $1/\eta$ queries), ensuring consistent noise patterns across all models.

This allows realistic simulation of real-life constraints.

4.4. Compared Models

Three models are evaluated under the same conditions.

The entropy model selects queries via information gain and uses a rigorous hard elimination. While the ML-based model selects queries using a pretrained XGBoost model with probabilistic belief updates. The proposed hybrid model, on the other hand, uses exploratory vs exploitation component with integrated XG boost model and incorporates heuristic components that ensure a dynamic approach and adapt itself based on the system's confidence.

All models share the same stopping criteria and simulation settings.

4.5. Evaluation Metrics

Performance is measured using accuracy, average number of questions, effective question cost, convergence rate, and robustness under noise.

4.6. Implementation Details

The system is implemented in Python using NumPy and Pandas for data processing and XGBoost for the machine learning component. Simulations are executed with fixed random seeds to ensure reproducibility.

5. Results and Analysis

A. Analysis under idea conditions (0% noise)

Under deal condition without noise, it is observed that the entropy model performs the best with an average of 11.1 questions. The ML model and adaptive model have roughly equal performance with 17.2 and 18.2 average questions, respectively the ML model edging slightly due to the model performance. The entropy also has a considerably better total score with 88.9 compared to the 82.8 and 81.8 for the ML model and Adaptive model, respectively.

The narrow spread in entropy indicates **deterministic convergence**, meaning the questioning strategy is consistently optimal regardless of the target.

In contrast, both ML and Adaptive models exhibit **higher variance**, implying that:

- Their decision-making depends on **intermediate probability distributions**
- The path to convergence is **data-dependent rather than structurally optimal**

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Table 5.1

Metrics	Entropy (Baseline1)	ML Engine (Baseline 2)	Adaptive engine
Total runs	100	100	100
Correct predictions	100	100	100
Accuracy %	100	100	100
Avg questions	11.1	17.2	18.2
Effective avgQ	11.1	17.2	18.2
TotalScore	88.9	82.8	81.8
Min questions	11	13	16
Max questions	12	24	22

Table 5.1 shows performance metrics under 0% noise

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B. Analysis under 5% noise

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- Their decision-making depends on **intermediate probability distributions**
- The path to convergence is **data-dependent rather than structurally optimal**

The narrow spread in entropy indicates **deterministic convergence**, meaning the questioning strategy is consistently optimal regardless of the target.

In contrast, both ML and Adaptive models exhibit **higher variance**, implying that:

Table 5.2

Metrics	Entropy (Baseline1)	ML Engine (Baseline 2)	Adaptive engine
Total runs	100	100	100
Correct predictions	34	99	99
Accuracy %	34	99	99
Avg questions	11.1	21.6	21.2
Effective avgQ	37.4	21.4	21.0
TotalScore	39.1	87.8	88.8
Min questions	11	13	16
Max questions	12	29	29

Table 5.2 shows performance metrics under 0% noise

- Their decision-making depends on **intermediate probability distributions**
- The path to convergence is **data-dependent rather than structurally optimal**

C. Analysis under 10% Noise

With the introduction of noise, the entropy model begins to break down, as reflected by the drop to **0% accuracy**. This can be attributed to its usage of hard elimination, where the correct solution might be discarded due to one or two incorrect answers by the users. Thus upon introduction of noise, the first approach completely collapses. Once removed, there is no mechanism for reintroducing it, leaving no scope for **partial or probabilistic recovery**.

- Despite this failure, the model continues to operate within a tight query range (**10–12 questions**).
 - This shows that the questioning strategy itself remains same.
 - However, convergence occurs toward incorrect outcomes, indicating **error propagation rather than inefficiency**. That is, the system doesn't know it has lost the actual result candidate along the process until the end and believes it's correct.

On the other hand, both the ML and adaptive models employ a **probability distribution-based approach**, ensuring that candidates are not prematurely eliminated.

- **Performance Comparison**
 - ML Engine: **~82–83% accuracy**
 - Adaptive Engine: **~92–93% accuracy**

Table 5.3

Metrics	Entropy (Baseline1)	ML Engine (Baseline 2)	Adaptive engine
Total runs	100	100	100
Correct predictions	0	83	92
Accuracy %	0	83	92
Avg questions	11.1	22.4	24.1
Effective avgQ	0	19.0	22.9
TotalScore	0.0	67.6	74.5
Min questions	10	15	16
Max questions	12	30	30

Table 5.3 shows performance metrics under 10% noise

- The adaptive model shows a clear advantage (~9–10% **higher accuracy**), which appears to stem from a more effective question selection strategy.

- Questioning Strategy Difference**

- The ML model leans towards learned patterns, typically favouring **balanced (near 50:50) splits** based on training data.
- The adaptive model, on the other hand, incorporates both:
 - Exploration** (reducing uncertainty across candidates)
 - Exploitation** (refining high-probability candidates)
- This allows it to adjust dynamically based on the evolving candidate space.

- Efficiency vs Robustness**

- The adaptive model requires only ~1.2–2 **additional questions on average** compared to ML.
- This slight increase results in a **significant gain in robustness and reliability**, suggesting that the additional queries are effectively utilised rather than redundant.

- The **min–max ranges (ML: 15–30, Adaptive: 16–30)** further indicate that both models operate under increased variability due to noise.

- However, the adaptive model achieves higher accuracy within a similar range, implying **better handling of uncertainty rather than increased search effort**.

- Additional Observation**

- These results are obtained under relatively **lenient exit conditions**, allowing earlier termination.
- Under stricter exit criteria, it is reasonable to expect further improvements in accuracy, particularly for the probabilistic models.

- Conclusion**

- ☐ **Entropy Model** → irreversible elimination → **no recovery**
- ☐ **ML Model** → probabilistic retention → **recoverable but inefficient**
- ☐ **Adaptive Model** → dynamic balancing of exploration and exploitation → **recoverable and efficient**

Key Observations

- Sensitivity to Noise**

- The entropy model undergoes an abrupt transition from perfect accuracy (100%) to complete failure (0%) upon the introduction of noise.
- Such a discontinuity is unlikely to reflect gradual degradation; rather, it points to an inherent fragility to input perturbations.
- By comparison, both ML and adaptive models display a more progressive decline, which ostensibly indicates greater tolerance to uncertainty.

- Query Growth Behaviour**

- The entropy model maintains a near-constant query count (~11) across conditions, suggesting a rigid and invariant questioning strategy.
- In contrast, the ML and adaptive models exhibit a ~30% increase in query count, implying an active, albeit differing, response to uncertainty.
- Notably, this additional query effort does not translate uniformly into performance gains, a discrepancy that warrants attention.

- Accuracy vs Cost Trade-off**

- Entropy retains optimal efficiency only in the absence of noise, yet fails to preserve correctness once perturbations are introduced.
- The ML model, while more stable, incurs a higher query cost, reflecting a trade-off that appears weighted toward reliability over efficiency.
- The adaptive model occupies an intermediate position, maintaining comparatively high accuracy with moderated query growth, thereby suggesting a more balanced allocation of computational effort.

- **Structural Interpretation**

- These patterns can be traced to the underlying model architectures:
 - Hard elimination (Entropy) — yields efficiency, but at the expense of recoverability
 - Static probabilistic reasoning (ML) — provides stability, though with limited adaptability
 - Dynamic hybrid strategy (Adaptive) — integrates exploration and exploitation, enabling both recovery and responsiveness

Conclusion:

Taken as whole, the results show that performance cannot be judged by pure efficiency alone in ideal conditions. Rather, the system's ability to accommodate and handle errors while inputs and its performance under noise must also be taken into account. While the entropy model shows an advantage and better results in ideal conditions, it irrevocably collapses upon the introduction of noise. The ML model, meanwhile, offers some robustness though with increased computation. The adaptive model, on the other hand, brings the best out of both worlds, showing a more consistent and context-sensitive performance while also ensuring better balance between accuracy, efficiency and resilience, thus presenting itself as the most viable approach in real-life conditions.

6. CONCLUSION

The system presented is an interactive song identification framework similar to the famous guessing game Akenator that integrates multiple questioning strategies and analyses them through multiple levels of uncertainty. By contrasting and evaluating the approaches within a simulated environment, we are able to not only get the outcomes but also the understanding behind the various approaches used and their efficiency.

The results thus show that efficiency alone is not enough to judge a method's effectiveness. While the entropy model achieves the fastest convergence, its hard elimination nature results in its collapse under the slightest of errors. In contrast, the other two approaches show greater stability and rigidity when introduced to ambiguities, though they exert more computational work. The adaptive model incorporates both exploratory and exploitative decision-making, appears to solve the problem, maintaining both a good accuracy and a lower query count at the same time.

It can be observed that simple strategy rigid elimination systems are handicapped in uncertain environments. Models that ensure flexibility, especially ones that can adjust their

questioning strategy based on the conditions are better suited to handle real-world interaction scenarios, where noise is unavoidable.

It must also be noted that the current evaluation and experimentation is done within a controlled simulation setting and certain assumptions still remain. Usage of lenient exit conditions, even by introducing noise, may not fully capture the variance of real-life user behaviour. Future work can therefore explore deployment of these strategies in live environments, showing richer feature representation, refine the strategies and include live user interaction through web hosting to ensure better framework, interaction and system design.

In conclusion the study showcases the importance of an adaptive uncertainty-aware system and the importance of dynamic approaches that can handle noise effectively. Rather than optimising just for minimal question count, a more balanced approach that accounts both for information gain, error handling and robustness under noise is essential for achieving reliable performance in practice.

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